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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

IOT PLATFORM DESIGN

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART – A	**5x2=10**		
1.	a.	Define Internet of Things (IoT).	L1	CO1	**02**
	b.	List two differences between RFID and NFC technologies.	L1	CO1	**02**
	c.	What is the significance of low power embedded systems in IoT?	L1	CO2	**02**
	d.	Explain the role of SPI protocol in sensor interfacing.	L1	CO2	**02**
	e.	What are the applications you can develop using Yoctoplatform?	L1	CO5	**02**
		PART - B	**6x5=30**		
2.	a.	Describe the transition of embedded systems into IoT.	L1	CO1	**05**
		OR			
	b.	Explain the architecture of an embedded system highlighting key hardware and software components.	L3	CO1	**05**
3.	a.	Discuss the significance of energy harvesting in IoT devices.	L2	CO2	**05**
		OR			
	b.	Write a short note on the power management techniques used in embedded systems.	L2	CO2	**05**
4.	a.	Describe how performance estimation is carried out for throughput in an embedded system.	L2	CO3	**05**
		OR			
	b.	Explain how debugging is integrated in embedded operating systems for IoT devices.	L4	CO3	**05**
5.	a.	How does the Yocto Project support the development of Linux-based embedded systems?	L2	CO4	**05**
		OR			
	b.	Illustrate the use of Yocto in a healthcare-based IoT system.	L3	CO4	**05**
6.	a.	Explain the importance of MAC and routing layers in embedded network protocols.	L2	CO5	**05**
		OR			
	b.	What are the key challenges in interfacing very low power sensors with embedded systems?	L2	CO5	**05**
7.	a.	Discuss the different debugging tools used in embedded systems for IoT applications.	L2	CO5	**05**
		OR			
	b.	Explain how optimization is achieved in embedded system coding.	L4	CO5	**05**

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		PART - C	6x10=60		
8.	a.	Analyze the challenges involved in partitioning hardware and software in complex IoT systems.	L5	CO1	10
		OR			
	b.	Design an algorithm for optimizing speed and energy efficiency in an IoT system with multiple sensors. Discuss its computational complexity and feasibility in embedded systems.	L6	C01	10
9.	a.	Evaluate different energy harvesting techniques for IoT applications. Compare their efficiency in terms of scalability, cost, and power output for low-power embedded devices.	L4	CO2	10
		OR			
	b.	Propose a power management solution for an IoT system that operates in an energy-constrained environment.	L6	CO2	10
10.	a.	Formulate a strategy for ensuring end-to-end code validation and robustness in IoT systems with distributed nodes.	L6	CO3	10
		OR			
	b.	Critically analyze the impact of system integration challenges in large-scale IoT deployments. How can debugging tools help mitigate these issues? Provide examples of real-world systems.	L5	CO3	10
11.	a.	Develop a case study on how the Yocto Project was used to create a custom Linux distribution for a high-stakes IoT application (e.g., autonomous vehicles or healthcare).	L6	CO4	10
		OR			
	b.	Compare and contrast different Linux-based distributions used in IoT systems in terms of performance, scalability, and support for embedded applications.	L5	CO4	10
12.	a.	Propose a novel architecture for integrating multiple embedded operating systems in a unified IoT network.	L6	CO5	10
		OR			
	b.	Evaluate the role of advanced debugging tools in improving the reliability and efficiency of IoT systems operating in unpredictable environments.	L5	CO5	10
13.	a.	Design an RFID-based IoT solution for supply chain management in a global company.	L6	CO5	10
		OR			
	b.	Critically assess the potential of NFC technology in enhancing IoT security features.	L5	CO5	10